

# Chat Session 2 — Hardware Architecture, Camera Decisions & Component Procurement

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**Date:** March 7, 2026

**Scope:** Hardware audit, camera placement architecture, wiring validation, component shopping, and camera module evaluation

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## 1. Session Overview

This session focused entirely on the **hardware side** of the Dual-Mode Vision-Guided Robotic Arm project. No code was written. The key outcomes were:

- Final camera architecture locked (3-camera setup)
  - Servo drooping behaviour understood and mitigated
  - Complete shopping list finalised with live product links
  - Wiring diagram validated for PSU → PCA9685 → ESP8266
  - OV7670 camera module evaluated and rejected
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## 2. Topics Discussed

### 2.1 Servo Drooping Under Load (MG996R)

**Problem raised:** MG996R servos were observed to droop/fall under load after the arm was positioned.

**Root cause analysis:**

- MG996R is open-loop — no internal position encoder or feedback
- If PWM signal is removed or goes idle, the servo releases its hold completely
- Under gravity load (especially at full horizontal extension), cumulative torque exceeds the rated 10 kg-cm stall torque

**Mitigations decided:**

| Mitigation                                          | Reasoning                                              |
|-----------------------------------------------------|--------------------------------------------------------|
| Always send hold PWM (never idle the signal)        | Servo holds position only when receiving active signal |
| Keep workspace within 15 cm of base                 | Reduces moment arm, reduces required torque            |
| Solid 5V / 10A PSU (no USB power)                   | PSU brownout causes servo jitter and positional drift  |
| Avoid full horizontal extension in workspace design | Worst-case torque scenario for shoulder joint          |

**Conclusion:** Drooping is a design constraint, not a fault. Work within servo limits.

## 2.2 IMU Module — MPU6050

**Decision:** Add MPU6050 to the wrist link for Phase 5 hardware deployment.

**Purpose:**

- Detect cumulative drift in wrist/end-effector orientation
- Cross-validate servo position estimates
- Used for drift correction, not full closed-loop control

**Specs confirmed:**

- I2C address: 0x68 (default, ADO pin low)
- Shares the same I2C bus as PCA9685 (no conflict — different addresses)
- Mounts physically on the wrist link of the arm
- Source: indianhobbycenter.com — ₹199

## 2.3 Camera Architecture — Final Decision

**The conflict:** User's phone was originally planned for both fixed overhead camera (YOLOv8) AND as the remote MQTT operator device.

**Resolution reached:** These are two completely different roles:

- User's phone = **always fixed overhead**, mounted on a stand above the workspace
- Remote operators = **their own phones**, accessing the ngrok web app URL from anywhere

**Final 3-Camera Architecture:**

| Camera           | Mount                | Purpose                         | Technology                                                                 |
|------------------|----------------------|---------------------------------|----------------------------------------------------------------------------|
| User's phone     | Fixed overhead stand | YOLOv8 object detection         | IP Webcam app → MJPEG stream → <code>cv2.VideoCapture("http://...")</code> |
| Laptop webcam    | Laptop lid           | LOCAL MediaPipe gesture control | <code>usb_cam</code> ROS2 node                                             |
| Any remote phone | Anywhere in world    | REMOTE MQTT gesture control     | Browser → MediaPipe.js → MQTT                                              |

**Why phone beats ESP32-CAM for fixed overhead:**

| Criterion  | Phone (IP Webcam) | ESP32-CAM   |
|------------|-------------------|-------------|
| Resolution | 1080p+            | 640×480 max |

| Criterion        | Phone (IP Webcam)                       | ESP32-CAM                      |
|------------------|-----------------------------------------|--------------------------------|
| Setup            | Install app, share URL                  | Flash firmware, configure WiFi |
| Cost             | ₹0 (already owned)                      | ~₹400                          |
| ROS2 integration | <code>cv2.VideoCapture</code> (2 lines) | Custom driver needed           |
| Reliability      | Proven MJPEG streaming                  | Known stability issues         |

**Winner: Phone (IP Webcam app) — zero cost, higher quality, easier integration.**

## 2.4 OV7670 Camera Module — Evaluated and Rejected

**Question:** Can OV7670 be used in this project?

**Analysis:**

| Issue                   | Detail                                                                                                                    |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------|
| No USB / network output | Raw parallel bus (D0–D7 + PCLK/VSYNC/HREF) — requires MCU to read; getting frames to ROS2 is a custom engineering problem |
| Resolution cap          | 640×480 VGA max — YOLOv8 accuracy drops significantly vs 1080p                                                            |
| No JPEG compression     | Raw RGB frame ≈ 900 KB; streaming over serial is unusably slow                                                            |
| Complex wiring          | 10+ GPIO lines with strict timing — notoriously unreliable even on Arduino                                                |
| No ROS2 driver          | No <code>usb_cam</code> or standard package supports it — full custom driver needed                                       |

**Verdict: Rejected.** The OV7670 is a microcontroller learning module. It is architecturally incompatible with a ROS2 + YOLOv8 pipeline. Phone + laptop webcam architecture is superior in every dimension.

## 3. Component Shopping List (Finalised)

### 3.1 From indianhobbycenter.com (Verified Live Links)

| Component                  | Link                                                                                                                                                                                                          | Price | Purpose                 |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------------------|
| MPU6050 6-axis IMU         | <a href="https://www.indianhobbycenter.com/products/mpu6050-6-axis-accelerometer-and-gyroscope-sensor">https://www.indianhobbycenter.com/products/mpu6050-6-axis-accelerometer-and-gyroscope-sensor</a>       | ₹199  | Wrist drift correction  |
| MB102 830-point Breadboard | <a href="https://www.indianhobbycenter.com/products/mb102-830-points-solderless-prototype-pcb-breadboard">https://www.indianhobbycenter.com/products/mb102-830-points-solderless-prototype-pcb-breadboard</a> | ₹75   | Prototyping connections |

| Component            | Link                                                                                                                                                                | Price | Purpose               |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----------------------|
| 400-point Breadboard | <a href="https://www.indianhobbycenter.com/products/small-breadboard-400-pin-copper">https://www.indianhobbycenter.com/products/small-breadboard-400-pin-copper</a> | ₹55   | Secondary prototyping |

**Note:** Jumper wires (40-pin Dupont packs) were NOT found on indianhobbycenter.com. Source from Amazon.in or Robu.in — search "40 pin dupont jumper wire male female" (~₹80–120).

### 3.2 Hardware Already On Hand

| Component                                   | Status            |
|---------------------------------------------|-------------------|
| 6-DOF Metal Servo Arm Kit                   | ✓ Received        |
| 6× MG996R Servo                             | ✓ Received        |
| ESP8266 NodeMCU                             | ✓ Received        |
| PCA9685 16-channel Servo Driver             | ✓ Received        |
| 5V / 10A PSU                                | ✓ Received        |
| 1080p Webcam                                | ✓ Received        |
| DC Power Jack Female (2-pin screw terminal) | ✓ Being purchased |

### 3.3 Final Remaining Items (Non-Electronic)

| Item                          | Why Needed                                                         | Source                    |
|-------------------------------|--------------------------------------------------------------------|---------------------------|
| Micro-USB cable               | Flashing firmware to ESP8266                                       | Any store / already owned |
| 2 mm flathead screwdriver     | <b>Critical</b> — tightening PCA9685 screw terminal block (V+/GND) | Hardware store            |
| Electrical tape / heat shrink | Insulating bare PSU wire joints                                    | Hardware store            |
| Zip ties                      | Cable management on arm                                            | Any store                 |
| Stable base / clamp for arm   | Prevent arm from tipping under load                                | Workshop / custom         |
| Phone stand / clamp           | Mount phone overhead above workspace                               | Amazon ~₹150–300          |

#### Confirmed NOT needed:

- Pull-up resistors for I2C → already on PCA9685 and MPU6050 boards
- Logic level converter → ESP8266 is 3.3V, PCA9685 accepts 3.3V on VCC/SDA/SCL
- Extra capacitors → only needed if PSU rail shows oscilloscope noise
- Additional MCU → ESP8266 is sufficient for this architecture

## 4. Wiring Architecture (Validated)

### 4.1 Power Domain Separation (Critical)

SERVO POWER DOMAIN:  
PSU (5V rail) → DC Jack Female (barrel) → Screw Terminal  
└ Red wire → PCA9685 V+  
└ Black wire → PCA9685 GND  
  
LOGIC POWER DOMAIN:  
ESP8266 3.3V → PCA9685 VCC (only powers the IC logic, NOT servos)  
ESP8266 GND → PCA9685 GND (MANDATORY – common ground reference)

**Critical rule:** VCC and V+ on PCA9685 are separate pins. VCC = logic (3.3V from ESP8266). V+ = servo power (5V from PSU). Never connect PSU directly to VCC.

### 4.2 I2C Bus Connections

ESP8266 D1 (GPIO5) → PCA9685 SCL  
ESP8266 D2 (GPIO4) → PCA9685 SDA  
  
MPU6050 SCL → same SCL line (address 0x68)  
MPU6050 SDA → same SDA line  
MPU6050 VCC → ESP8266 3.3V  
MPU6050 GND → Common GND

### 4.3 Wire Gauge

| Connection                         | Wire Gauge                |
|------------------------------------|---------------------------|
| PSU → PCA9685 V+/GND (servo power) | 20–22 AWG                 |
| ESP8266 → PCA9685 VCC/GND (logic)  | 24–26 AWG (any thin wire) |
| I2C lines (SCL/SDA)                | 24–26 AWG                 |

### 4.4 DC Power Jack Female Connector (Screw Terminal Type)

**Confirmation:** DC Power Jack Female with 2-pin PBT screw terminal is the **correct connector** for connecting the PSU barrel output to the PCA9685 power input.

- Centre pin = positive (V+) → red wire → PCA9685 V+
- Outer sleeve = negative (GND) → black wire → PCA9685 GND
- Screw down firmly with 2 mm flathead screwdriver

## 5. Architecture Approaches Considered & Decisions Made

### 5.1 Fixed Camera — Approaches Considered

| Approach                    | Considered | Rejected Reason                                                      |
|-----------------------------|------------|----------------------------------------------------------------------|
| ESP32-CAM module            | Yes        | Lower resolution, firmware complexity, ~₹400 cost                    |
| OV7670 raw sensor           | Yes        | No network output, parallel bus incompatible with ROS2               |
| USB webcam mounted overhead | Yes        | Cable management difficult, webcam already used for gesture          |
| Phone (IP Webcam app)       | Yes        | <b>CHOSEN — 1080p, free, MJPEG stream, zero integration friction</b> |

### 5.2 Servo Controller — Approaches Considered

| Approach                     | Considered | Rejected Reason                                                       |
|------------------------------|------------|-----------------------------------------------------------------------|
| Arduino Mega (original plan) | Yes        | USB tether, no wireless, limited for future phases                    |
| Raspberry Pi direct GPIO     | Yes        | 3.3V PWM output, RPi GPIO not designed for servo load                 |
| ESP8266 + PCA9685            | Yes        | <b>CHOSEN — Wi-Fi capable, I2C clean, 16-channel, 5V servo signal</b> |

### 5.3 Remote Control Infrastructure — Approaches Considered

| Approach                   | Considered | Rejected Reason                                                    |
|----------------------------|------------|--------------------------------------------------------------------|
| Direct WebSocket to laptop | Considered | Requires port forwarding, fails behind NAT/firewall                |
| WebRTC                     | Considered | Complex setup, overkill for joint angle streaming                  |
| Custom TCP server + VPN    | Considered | Requires VPN setup, not phone-accessible                           |
| MQTT + HiveMQ + ngrok      | Yes        | <b>CHOSEN — free, NAT-penetrating, reliable on mobile networks</b> |

## 6. Project Phase Status (as of this session)

| Phase   | Description                            | Status                                                                                       |
|---------|----------------------------------------|----------------------------------------------------------------------------------------------|
| Phase 0 | URDF, Gazebo simulation, RViz2 TF tree |  Complete |

| Phase     | Description                                       | Status                                          |
|-----------|---------------------------------------------------|-------------------------------------------------|
| Pre-Phase | MediaPipe hand detection pipeline                 | ✅ Verified working                              |
| Phase 1   | Movelt2 Setup Assistant + SRDF                    | ⌚ Not started — <b>immediate next step</b>      |
| Phase 2   | Gesture → arm joint mapping (live simulation)     | 🔄 Partial (landmarks work, mapping not written) |
| Phase 3   | YOLOv8 autonomous sorting pipeline                | ⌚ Not started                                   |
| Phase 4   | MQTT remote control (Flask + ngrok + HiveMQ)      | ⌚ Not started                                   |
| Phase 5   | Hardware deployment (ESP8266 + PCA9685 + MPU6050) | ⌚ Not started                                   |

## 7. Immediate Next Actions

### 1. Order remaining components:

- MPU6050 from indianhobbycenter.com (₹199)
- 40-pin Dupont jumper wires from Amazon/Robu.in (~₹100)
- Phone overhead stand (~₹150–300)
- 2 mm flathead screwdriver

### 2. Begin Phase 1 — Movelt2 Setup:

```
sudo apt install ros-jazzy-moveit
ros2 launch moveit_setup_assistant setup_assistant.launch.py
```

- Load: **robot\_arm\_description** URDF
- Configure planning group: **arm** (joint\_1 → joint\_5)
- Configure end-effector: **gripper** (parent: gripper\_base link)
- Generate self-collision matrix (SRDF)
- Export **robot\_arm\_moveit\_config** package

### 3. Test Movelt2: Drag interactive marker in RViz2 → Plan and Execute → verify arm moves in simulation

## 8. Key Technical Decisions Summary

| Decision              | Choice Made           | Rationale                                      |
|-----------------------|-----------------------|------------------------------------------------|
| Fixed overhead camera | Phone (IP Webcam app) | Free, 1080p, MJPEG, works with OpenCV directly |

| Decision               | Choice Made                  | Rationale                                           |
|------------------------|------------------------------|-----------------------------------------------------|
| Remote operator camera | Any phone (browser)          | MediaPipe.js runs in browser, no app install needed |
| IMU for drift feedback | MPU6050 on wrist link        | Low cost, I2C compatible, shares ESP8266 bus        |
| Servo controller       | ESP8266 + PCA9685            | Wi-Fi ready, I2C clean, 16 channels                 |
| Power supply           | Dedicated 5V/10A PSU         | Prevents brownout; USB/laptop power insufficient    |
| ROS middleware         | ROS2 Jazzy + Gazebo Harmonic | Ubuntu 24.04 compatible pair, active LTS            |
| Motion planning        | MovelIt2 + OMPL              | Industry standard, collision-aware IK               |
| Object detection       | YOLOv8 (CUDA) on RTX 3060    | ~15ms inference, >90% accuracy on standard objects  |

*Session ended with hardware architecture fully resolved. No code changes made. Next session should begin Phase 1 (MovelIt2).*